

COMP 3311: Database Management Systems

Relational Database Creation and SQL Queries Assignment

Assigned: July 2, 2026

Value: 10%

Due: 11:00 p.m., July 11, 2026

This assignment requires you to submit two SQL script files that create the relational database required for the movie streaming management system and that execute some queries on this database.

IMPORTANT REMINDER

This is an individual assignment. The SQL script files that you submit should be *your own work*. While you may discuss general assignment issues with other students, you are not allowed to collaborate with any person or AI agent to arrive at a solution, to share solutions or to copy someone else's solution. Collaborating, sharing and copying will be severely penalized. All those involved in a collaborating/sharing/copying case will automatically receive a grade of 0 and may be reported for further disciplinary action.

PART 1: RELATIONAL DATABASE CREATION

In the script file **sqlCreatedb.sql** construct SQL DDL statements that create tables with exactly the same table and attribute names as those of the relation schemas in the document **Relation Schemas**. Moreover, the order of the attributes and their types in each table should be exactly the same as that of the corresponding relation schema in the document **Relation Schemas**. Note that the order of the relation schemas in this document is ***not necessarily the correct order*** for creating the tables; the relation schemas in this document merely describe the attributes, their types and some of the constraints of each relation schema.

Specify for each table all attribute-level or table-level constraints that either are described in the application description, are derivable from the example E-R schema for the *Entity-Relationship Schema Design* assignment or are stated in the document **Relation Schemas**.

Checking Table Definitions

You can check whether you have defined *some aspects* of your tables correctly with the script file **checkTables.sql**. If each tuple in this script file can be successfully inserted into its respective table, then the table has been defined correctly as to the number of the attributes and possibly the order and type of each attribute. However, the order and type of the attributes is not guaranteed to be correct even if a tuple can be inserted successfully. Furthermore, the tuples need to be inserted in the correct order and the order in which they appear in the script file is **not** necessarily the correct order in which the tables should be created. The tuples may need to be reordered according to any specified foreign key (referential integrity) constraints so that they can be successfully inserted into the tables.

PART 2: SQL QUERIES

In the SQL script file **sqlQueries.sql** construct SQL DML queries in the indicated places that retrieve the required information as specified by the query description. You must use the relation schemas in the document **Relation Schemas** to construct your queries.

IMPORTANT NOTE

You are allowed to use only SQL constructs that have been discussed in the course. If SQL constructs that have not been discussed in the course are used in a query, then no marks will be given for that query.

Do not use PL/SQL procedures or functions; use SQL statements only.

Example Query Results

Below are example query results for each query illustrating how the query result should be presented.

Query 1:

TITLE	GENRES
Sunset Boulevard	dark romance, film-noir

Query 2:

NAME	ROLE	TITLE	RELEASEYEAR
Mel Gibson	William Wallace	Braveheart	1995

Query 3:

NAME	MOVIES
Ralph Fiennes	(Schindler's List, 1993; The Hurt Locker, 2008; The English Patient, 1996)

Query 4:

TITLE	IMDBRATING	REELFLICSRATING
Hara-Kiri	8.6	9

Query 5:

TITLE	RELEASEYEAR	DIRECTOR	ACTOR	ACTRESS
It Happened One Night	1934	Frank Capra	Clark Gable	Claudette Colbert

Query 6:

TITLE	MPAARATING	IMDBRATING	REELFLICSRATING	TIMESSTREAMED
Amadeus	PG	8.3	7.3	3

ASSIGNMENT FILES

All files required for this assignment are in the zipped folder **Relational Database Creation and SQL Queries Files** which can be downloaded from the entry for this assignment on the Canvas *Assignments* page.

Note: *You are required to use the two script files **sqlCreatedb.sql** and **sqlQueries.sql** as well as the relation schemas described in the document **Relation Schemas**. For this assignment we will not provide any data to populate the database. You may create your own data if you consider it is necessary to do so.*

WHAT TO SUBMIT

Submit your two completed SQL script files **sqlCreatedb.sql** and **sqlQueries.sql**.

You must use and submit these two script files. No other type of file submission is acceptable.

Important Notes

1. The file **sqlCreatedb.sql** should not contain any statements that insert data; it should contain only the SQL DDL statements for creating the relations.
2. Your name and student id should be included in the indicated place in each script file.
3. We will not grade your submission if
 - your **sqlCreatedb.sql** script file violates any of the requirements stated in Part 1,
 - you do not use the two provided script files **sqlCreatedb.sql** and **sqlQueries.sql** or
 - you do not put your name and student id in each script file correctly.

Note: *Each script file will be run as a script using Oracle SQL Developer. If a script file cannot be executed, it will not be graded, and you will receive 0 for that part of this assignment.*

PLEASE NOTE CAREFULLY

You may not share your SQL script files for Part 1 or Part 2 with other students in the course. Sharing your script files will be considered collaboration and will be penalized as stated above.

HOW TO SUBMIT

1. Place both script files into a folder named **sql<login id>** where **<login id>** is your HKUST login id.
2. Compress (zip) the folder **sql<login id>**.
3. **By 11:00 p.m. on Saturday, July 11**, upload your **sql<login id>.zip** file to the *Relational Database Creation and SQL Queries* assignment on Canvas.

It is your responsibility to ensure that your *sql<login id>.zip* file is correctly uploaded to Canvas.
Under no circumstances will late submission or submission by email be accepted.
Moreover, any submission to the assignment's comments section will be ignored.

GRADING

<u>Item</u>	<u>Value</u>
Part 1: Relational database creation Grading will be based on correct specification of the table creation order and specifying all required constraints for the relation schemas.	~50%
Part 2: SQL queries Grading of each query will be as follows: 0% no query is submitted, or it cannot be executed. 10% query raises error 25% query result is empty 50% query result contains none or only some of the expected result tuples 75% query result contains all expected tuples plus additional incorrect result tuples 90% query result contains only expected result tuples, but there are duplicate tuples and/or not all expected columns are present and/or tuples are not ordered or formatted as specified	~50%

CLARIFICATION AND AMENDMENT OF ASSIGNMENT REQUIREMENTS

Clarification questions about the assignment should be submitted to the teaching team by email at 3311rep@cse.ust.hk. In addition to being answered individually, if appropriate, a submitted question and its reply will be posted in the *Relational Database Creation and SQL Queries Q&A* document on the Canvas page for this assignment. ***This document should be consulted on a regular basis for further clarification and amendment of assignment requirements.*** Any requirements added or amended in this document become part of the assignment requirements.